

## INSPECTION REPORT

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[FICTIONAL/DEMO DATA - NOT REAL EQUIPMENT]

Organization: DEMO Refinery Corporation (FICTIONAL)  
Plant / Unit: Crude Distillation Unit (CDU) - DEMO  
Report No: IR-2024-1047 (FICTIONAL)  
Date of Inspection: 2024-11-15  
Inspector: J. Sharma (FICTIONAL)  
Witness: R. Verma (FICTIONAL)

### EQUIPMENT DETAILS

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Equipment Name: Crude Oil Transfer Pump  
Equipment ID / Tag: P-104  
Equipment Type: Centrifugal Pump  
Service: Crude Oil Transfer (FICTIONAL)  
Rated Flow: 250 m3/hr  
Rated Head: 45 m  
Motor Power: 55 kW  
Last Maintenance: 2024-06-10

### INSPECTION FINDINGS

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1. VIBRATION - ABNORMAL  
Vibration readings at pump discharge end: 8.7 mm/s (RMS)  
Acceptable limit: 4.5 mm/s  
Status: EXCEEDS LIMIT - Requires immediate attention
2. SEAL LEAKAGE - DETECTED  
Mechanical seal showing minor leakage at drive end  
Estimated leakage rate: 3-4 drops/minute  
Seal type: Single mechanical seal, Type A  
Status: DEGRADED - Schedule seal replacement
3. BEARING TEMPERATURE - ELEVATED  
Drive end bearing: 86 degrees Celsius  
Non-drive end bearing: 79 degrees Celsius  
Normal operating range: 55-75 degrees Celsius  
Status: ABOVE NORMAL - Monitor and schedule inspection
4. OPERATING PRESSURE  
Suction pressure: 2.1 bar  
Discharge pressure: 17.2 bar  
Differential pressure: 15.1 bar

Rated differential pressure: 16.0 bar  
Status: SLIGHTLY BELOW RATED - Acceptable

5. NOISE - ABNORMAL

Unusual metallic noise detected at non-drive end bearing  
Possible causes: Bearing wear, cavitation, misalignment  
Status: INVESTIGATE

6. COUPLING CONDITION

Flexible coupling shows signs of rubber element wear  
Estimated remaining life: 3-6 months  
Status: MONITOR

SUMMARY

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Equipment P-104 shows multiple concurrent issues requiring maintenance attention. The combination of elevated bearing temperature, abnormal vibration, and seal leakage suggests possible bearing wear or misalignment condition.

RECOMMENDATIONS

- 1. Immediate: Reduce load on pump and increase monitoring frequency  
2. Short-term (within 7 days): Replace mechanical seal  
3. Short-term (within 14 days): Inspect and replace bearings if confirmed worn  
4. Short-term: Re-align pump-motor coupling  
5. Long-term: Schedule complete pump overhaul during next planned shutdown

RISK ASSESSMENT

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Risk Level: HIGH  
Consequences if not addressed: Potential pump failure, process disruption, environmental release, safety hazard to personnel.

Inspector Signature: \_\_\_\_\_ Date: 2024-11-15  
Supervisor Review: \_\_\_\_\_ Date: \_\_\_\_\_

[THIS IS FICTIONAL DEMO DATA - ALL EQUIPMENT, MEASUREMENTS, AND FINDINGS ARE INVENTED]